Ex No:3	**Visualization Techniques for Qualitative Data**
Date	

Aim:

To analyze and apply various data visualization techniques for qualitative (categorical) data in order to effectively represent category-wise distributions and relationships.

Software Used:

• Programming Language: Python 3.x

• Libraries:

- o Pandas (for data loading and manipulation)
- o Matplotlib (for basic visualizations)
- o Seaborn (for enhanced categorical plots)
- o WordCloud (for textual visualization)

• Environment: Jupyter Notebook.

Description

Qualitative data represents categorical or non-numeric information such as names, labels, and classifications. Visualizing qualitative data helps in understanding frequency distributions, proportions, and relationships among categories.

In this experiment, a sample dataset related to social media platforms is analyzed using multiple qualitative visualization techniques. The dataset includes categorical attributes such as platform names, user types, and engagement levels.

The following visualization techniques are applied:

• **Bar Chart** – To compare the frequency of different social media platforms.

• **Pie Chart** – To represent the proportion of different user types.

• **Stacked Bar Chart** – To visualize engagement levels across platforms.

• **Word Cloud** – To highlight frequently occurring platform names visually.

These visualizations provide insights into platform popularity, user distribution, and engagement patterns.

Dataset details

The dataset contains qualitative attributes related to social media usage.

• Platform – Name of the social media platform (Facebook, Instagram, Twitter, LinkedIn, YouTube).

• User Type – Type of user (Business or Personal).

• Engagement Level – User engagement category (High, Medium, Low).

• Rows: 8 sample entries (simulated dataset for analysis).

List of Tasks

1. Import required Python libraries (Pandas, Matplotlib, Seaborn, WordCloud).

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud, STOPWORDS
```

2. Create a DataFrame containing qualitative social media data.

```
data=pd.read_csv("social_media_data.csv")
data.head()
data = {
    "Platform": [
        "Facebook", "Instagram", "Twitter", "LinkedIn",
        "YouTube", "Instagram", "Facebook", "Twitter"
    ],
    "User Type": [
        "Personal", "Personal", "Business", "Business",
        "Personal", "Business", "Business", "Personal"
    ],
    "Engagement Level": [
        "High", "High", "Medium", "Medium",
        "High", "Low", "Medium", "Low"
    ]
}
df = pd.DataFrame(data)
df
```

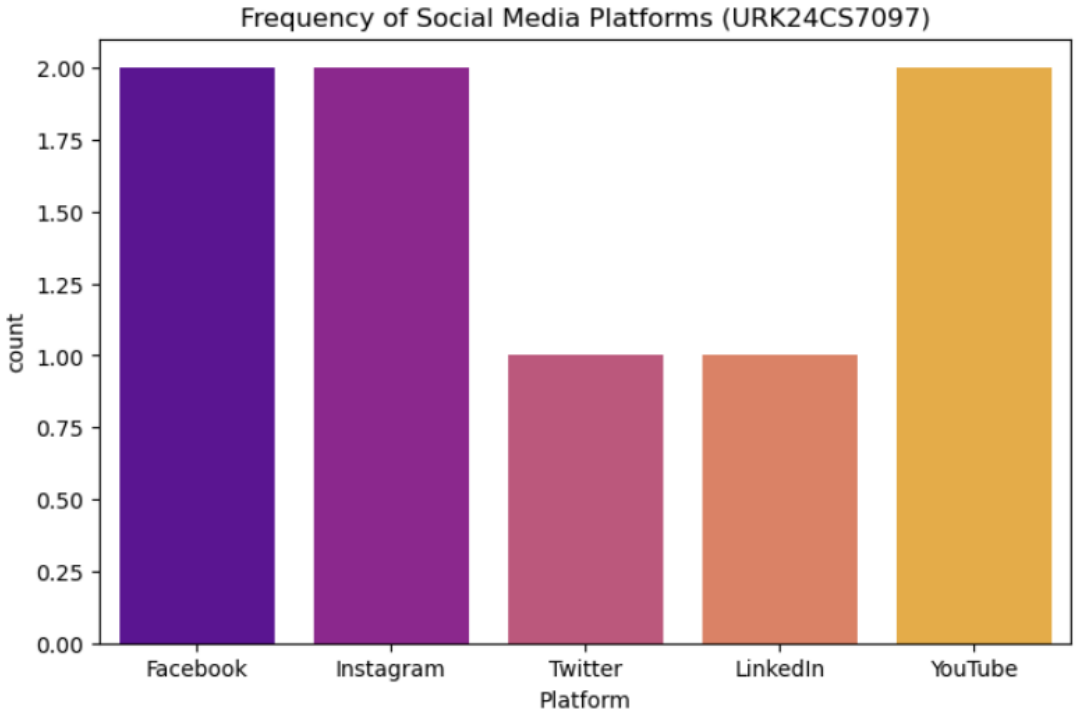
OUTPUT:

	Platform	User Type	Engagement Level
0	Facebook	Personal	High
1	Instagram	Personal	Low
2	Twitter	Personal	Medium
3	LinkedIn	Business	High
4	YouTube	Personal	Low
5	Instagram	Business	High
6	Facebook	Personal	Medium
7	YouTube	Business	High

3. Generate a Bar Chart to show the frequency of platforms.

```
plt.figure(figsize=(5, 5))
sns.countplot(data=df, x="Platform", palette="Set2")
plt.title("Frequency of Social Media Platforms-7081")
plt.xlabel("Platform")
plt.ylabel("Count")
plt.show()
```

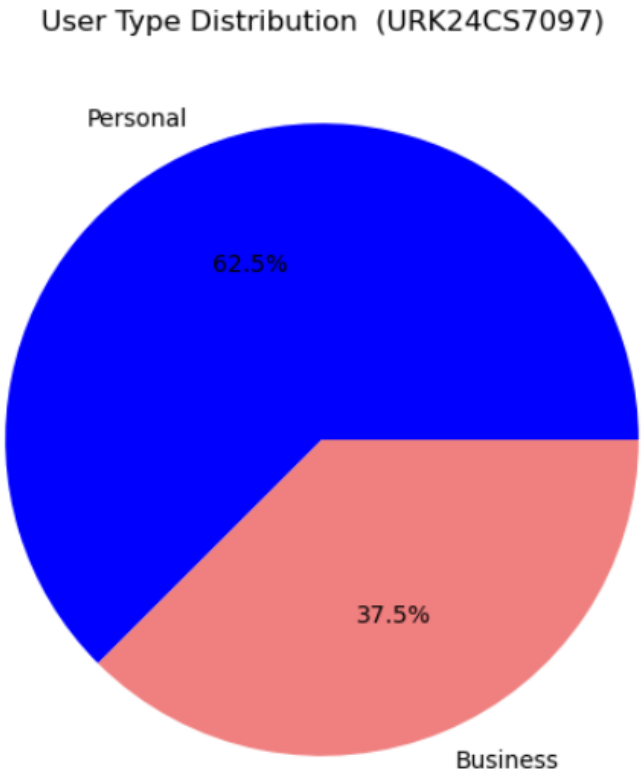
OUTPUT:



4. Create a Pie Chart to represent the proportion of user types.

```
user_type_counts = df["User Type"].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(user_type_counts,labels=user_type_counts.index,autopct="%1.1f%%",startangle=140,colors=["violet", "skyblue"])
plt.title("Proportion of User Types-7081")
plt.show()
```

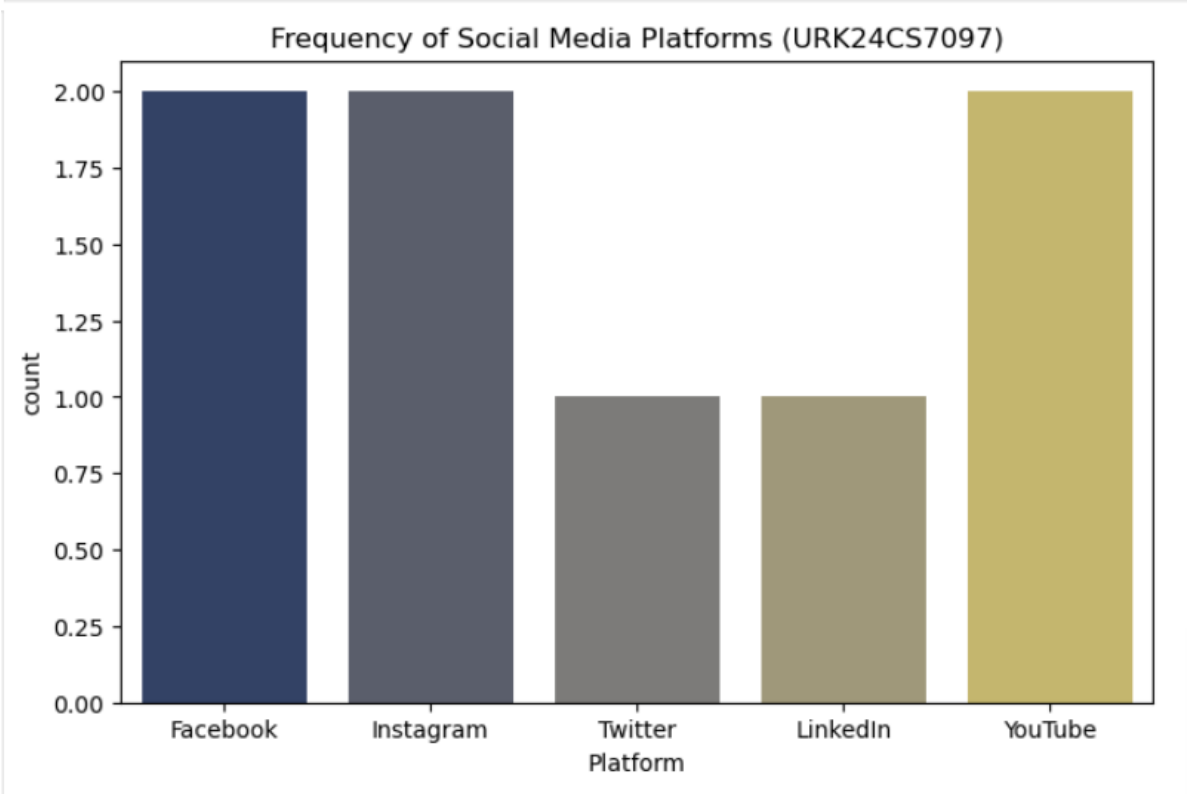
OUTPUT:



5. Draw a Stacked Bar Chart to visualize engagement levels by platform.

```
engagement_platform = pd.crosstab(df["Platform"], df["Engagement Level"])
engagement_platform.plot(kind="bar",stacked=True,figsize=(8, 5),color=["silver","beige","grey"])
plt.title("Engagement Levels Across Platforms")
plt.xlabel("Platform")
plt.ylabel("Number of Users")
plt.legend(title="Engagement Level")
plt.show()
```

OUTPUT:



6. Generate a Word Cloud to visualize frequently occurring platform names.

```
platform_text = " ".join(df["Platform"])
wordcloud =
WordCloud(width=300,height=300,background_color="white").generate(platform_text)
plt.figure(figsize=(5, 5))
plt.imshow(wordcloud)
plt.axis("off")
plt.title("Word Cloud of Social Media Platforms-7081")
plt.show()
```

OUTPUT:

7. Interpret the visualizations to identify patterns and insights.

Video Link –

https://youtu.be/FT8RI29Hg3k

RESULT: based on visualization technique for qualitative data the programs have been executed and outputs are executed successfully.